

SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA1201	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL3- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Answer

Cache memory is organized into multiple levels to reduce the average time required by the CPU to access frequently used data. L1, L2, and L3 differ mainly in size, latency, bandwidth, and proximity to the processor.

Feature	L1 Cache	L2 Cache	L3 Cache
Location	Closest to CPU core	Near/within CPU core	Usually shared among cores
Typical Size	16 KB–128 KB per core	256 KB–2 MB per core	Several MB to tens of MB
Latency	~1–4 cycles	~5–15 cycles	~20–50+ cycles
Bandwidth	Highest	High	Lower than L1/L2
Access Speed	Fastest	Very fast	Slower than L1/L2
Main Purpose	Immediate data/instruction access	Secondary fast storage	Shared data and reducing accesses

L1 cache provides the lowest latency and highest effective bandwidth because it is located closest to the execution units. It is usually divided into instruction and data caches. Its small capacity allows very fast access.

L2 cache is larger than L1 but has higher latency. It stores data that cannot fit in L1 and acts as an intermediate buffer before accessing lower-level memory.

L3 cache is significantly larger and is commonly shared between processor cores. Although it has higher latency and lower bandwidth than L1 and L2, it reduces expensive accesses to main memory.

Impact on processor performance

The hierarchy improves performance because most frequently accessed data can be found in the faster cache levels. A typical access sequence is:

CPU → L1 → L2 → L3 → DRAM

A high L1/L2 hit rate reduces the average memory access time. L3 is especially important in multicore processors because shared data can be accessed without immediately going to DRAM.

For example, if L1 access takes about 2 cycles while DRAM access takes roughly 100+ cycles, avoiding a DRAM access can save a large number of CPU cycles.

Critical analysis: Increasing cache size does not automatically increase performance. Larger caches generally have greater latency and hardware cost. Therefore, modern processors use a hierarchy that balances speed, capacity, bandwidth, and cost. This directly supports the CO4 objective of implementing hierarchical memory systems based on latency, bandwidth, and throughput.

2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

Answer

SRAM and DRAM are both volatile memory technologies, but their different internal structures make them suitable for different levels of the memory hierarchy.

Feature	SRAM	DRAM
Storage Cell	Flip-flop, typically 6 transistors	Capacitor + transistor
Latency	Very low	Higher
Bandwidth	Very high	High, but generally lower effective latency
Throughput	Very high for low-latency accesses	High for large sequential transfers
Refresh	Not required	Required periodically
Density	Lower	Higher
Cost per bit	Higher	Lower
Typical Use	CPU caches	Main memory

SRAM in cache design

SRAM is preferred for **L1, L2 and many L3 caches** because it can provide fast access without refresh operations. Its low latency allows the processor to retrieve frequently used instructions and data quickly. However, SRAM requires more transistors per bit and therefore occupies more chip area. This makes large SRAM caches expensive.

DRAM in main memory

DRAM stores each bit using a transistor and capacitor. Because the capacitor gradually loses its charge, DRAM requires periodic refreshing. This increases control complexity and contributes to greater access latency.

Its major advantage is high density and lower cost per bit, making it practical for large main-memory capacities.

Performance interpretation

SRAM is optimized for low latency and rapid repeated access, whereas DRAM is optimized for high capacity and economical storage.

For example:

CPU → SRAM Cache → DRAM Main Memory

If requested data is available in SRAM cache, the CPU avoids the considerably slower DRAM access.

Critical analysis: Using SRAM for all main memory would provide excellent latency but would be prohibitively expensive and area-intensive. Conversely, using DRAM for cache would increase processor stalls. Therefore, combining SRAM and DRAM provides an effective balance between latency, bandwidth, capacity, and cost.

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Answer

Virtual memory with paging and cache memory both improve system operation by managing different levels of the memory hierarchy, but they solve different problems.

Feature	Virtual Memory + Paging	Cache Memory
Purpose	Provides large logical address space	Reduces CPU memory-access late
Managed by	OS + MMU	Hardware/cache controller
Storage	RAM + secondary storage	SRAM-based cache
Access Latenc	Very low for RAM hits, extremely high durin faults	Very low
Data Unit	Pages	Cache lines/blocks
Miss Penalty	Very high for page fault	Much lower than disk/page-fault p

Feature	Virtual Memory + Paging	Cache Memory
Main Benefit	Memory capacity and process isolation	Speed and CPU performance

Cache mechanism

Cache memory keeps frequently or recently accessed data close to the CPU. When the required data is found, a **cache hit** occurs. A cache miss requires the processor to access a lower level of the memory hierarchy.

Therefore:

Cache hit → Low latency → Higher CPU throughput

Virtual memory and paging

Virtual memory allows programs to use an address space larger than the available physical RAM. The address space is divided into pages, while physical memory is divided into page frames.

When a required page is not present in RAM, a page fault occurs. The operating system must retrieve the page from secondary storage, producing a very large latency compared with cache or DRAM access.

Impact on throughput

Cache memory directly improves CPU throughput by reducing the time processors spend waiting for memory.

Virtual memory improves system-level throughput and flexibility by allowing multiple programs to execute while providing isolation and efficient memory allocation. However, excessive page faults can cause thrashing, where the system spends more time transferring pages than executing programs.

Critical analysis: Cache and virtual memory should not be viewed as competing technologies. They operate at different levels of the hierarchy:

CPU → Cache → DRAM → Virtual Memory/Storage

Cache primarily addresses latency, whereas virtual memory primarily addresses capacity, protection, and process management. An efficient system requires both mechanisms.

4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

Answer

NAND Flash and DRAM are both widely used memory technologies, but their characteristics are fundamentally different. DRAM is volatile and optimized for fast working memory, while NAND Flash is non-volatile and optimized for high-density storage.

Feature	DRAM	NAND Flash
Volatility	Volatile	Non-volatile

Feature	DRAM	NAND Flash
Latency	Low	Much higher
Bandwidth	High	High sequential bandwidth possible
Random Access	Very efficient	Relatively slow
Write Operation	Fast	Slower and requires program/erase operations
Endurance	Very high for normal operation	Limited program/erase cycles
Density	Lower	Very high
Cost per Bit	Higher	Lower
Typical Use	Main memory	SSDs, flash storage

DRAM

DRAM provides low-latency random access and is therefore suitable for main memory. The CPU can access individual memory locations efficiently, making DRAM appropriate for active programs and data.

NAND Flash

NAND Flash retains data even when power is removed. It provides high storage density and relatively low cost per bit. However, its access characteristics are much slower than DRAM, particularly for small random operations.

NAND Flash is organized around pages and erase blocks. Data writing and erasing involve additional management operations, which contribute to latency.

Suitability in hierarchical memory

A typical hierarchy is:

CPU → Cache → DRAM → NAND Flash

DRAM is positioned closer to the CPU because its lower latency makes it suitable for active computation. NAND Flash is positioned farther away because it offers much greater capacity and non-volatility.

Critical analysis: NAND Flash can provide impressive sequential throughput, especially in modern SSDs, but throughput should not be confused with low latency. DRAM remains much more suitable for workloads requiring frequent random accesses.

Thus, DRAM optimizes working-memory performance, while NAND Flash optimizes capacity, persistence, and cost efficiency.

5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Answer

MRAM (Magnetoresistive RAM) and **RRAM (Resistive RAM)** are emerging non-volatile memory technologies designed to overcome some limitations of conventional memory systems.

Feature	MRAM	RRAM
Storage Principle	Magnetic state	Resistance state
Volatility	Non-volatile	Non-volatile
Access Time	Very low	Potentially very low
Read Speed	Fast	Fast
Write Speed	Fast	Fast, technology-dependent
Throughput	High	High potential
Energy Efficiency	Good	Potentially very good
Endurance	Generally high	Depends strongly on material/device
Scalability	Good	Very high potential
Applications	Embedded memory, applications	High-density storage and neuromorphic computing

MRAM

MRAM stores information using magnetic states rather than electrical charge. Since it is non-volatile, it can retain data without continuous power.

Its major advantage is the combination of **fast access, non-volatility, and high endurance**. This makes MRAM attractive for embedded systems and memory applications requiring frequent read/write operations.

RRAM

RRAM stores information by changing the resistance of a material. It can potentially achieve high density because its cell structure can be compact.

RRAM is attractive for future high-density memory systems and specialized computing applications. Its energy consumption can be low, particularly for certain read operations, although write characteristics vary according to the implementation.

Performance comparison

MRAM generally offers a strong combination of speed, endurance, and non-volatility, while RRAM has significant potential for high density and energy-efficient operation.

Critical analysis: Neither technology universally replaces SRAM or DRAM. Their suitability depends on workload requirements. MRAM can be attractive where fast access and endurance are important, whereas RRAM can be particularly attractive where density and low-energy operation are priorities. Therefore, both technologies have potential to improve future hierarchical memory systems by providing a compromise between volatile high-speed memory and slower non-volatile storage.

Code of Conduct:

*I **MRUDHULA K (192524335)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MRUDHULA K

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided

Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand